

Sprint 2 Readiness

Project: Image Guided Surgery Intervention

Sprint 2 Goals

- 3D Volume Reconstruction: Implement the core algorithm to transform 2D ultrasound scans into 3D volumes in real-time.
- Database integration: Finalize the integration of the database to handle 2D and 3D scan storage and retrieval.
- Hardware Integration & Testing: Validate the software pipeline using the physical ultrasound scanner and positional tracker.
- Architecture Maintenance: Perform continuous code refactoring and generate comprehensive documentation for new modules to ensure extensibility.

Backlog Review

This table reviews the status of Sprint 1 work and introduces the new work planned for Sprint 2.

Story ID	Title	Status in Sprint 1	Actions for Sprint 2	Notes/Reason
S1-01	Familiarization with Existing Software	Completed	None	Team has reviewed existing repo and documentation.
S1-02	Setup and Documentation	Completed	None	Development environments are reproducible and setup procedure is documented.
S1-03	Modularization and Extensibility	Completed	None	Core components such as OverlayApp.py and OverlayRenderer.py, have been refactored, further refactoring moved to new continuous maintenance story.
S1-04	Database Integration	Completed	None	Database used to store patient scans integrated.
S1-05	Tests (Initial)	In Progress	Carry Over	Initial tests created, but coverage needs to increase.

S2-01	3D Reconstruction Algorithm	Not Started	New Story	Top Priority: Core feature for Sprint 2
S2-02	Database Scan Storage	Not Started	New Story	Backend requirement for saving scans, enhanced to store 3D scans.
S2-03	Ultrasound Hardware Integration	Not Started	New Story	Connecting physical scanner to software pipeline/data stream.
S2-04	Continuous Refactoring and Documentation	Not Started	New Story	Ongoing maintenance as new features are added.

Sprint Planning Details

Breakdown of tasks selected for Sprint 2, assigned based on the Roles/RACI chart in the Team Charter.

Story/Task	Assigned To	Est. Points	Status
S2-01: Implement 3D Reconstruction Algorithm	Delaney McLachlan	8	Ready
S2-02: Database Integration (Scan Storage)	Hannah Taylor	5	Ready
S2-03: Hardware Integration (Ultrasound Stream)	Muneeb Asif	8	Ready
S2-04: Continuous Code Refactoring	Elise Drouillard	3	Ready
S2-05: Expand Unit Test Coverage	Janki Patel	5	In Progress
S2-06: API & System Documentation	Elise Drouillard	3	Ready